



Figure 1: Docker on Cloud9

are extremely popular, that doesn't mean virtual machines are out of date. Which of the two is selected depends entirely on the specific needs of the end user.

Virtual machines (VMs) run on top of a hypervisor with a fully virtualised and totally isolated OS. They take up a lot of system resources and are also very slow to move around. Each VM runs not just a full copy of an operating system, but a virtual copy of all the hardware that the operating system needs to run. This quickly adds up to a lot of RAM and CPU cycles. And yes, containers can enable your company to pack a lot more applications into a single physical server than a VM can. Container technologies such as Docker beat VMs at this point in the cloud or data centre game.

Virtual machines are based on the concept of virtualisation, which is the emulation of computer hardware. It emulates hardware like the CPU, RAM and I/O devices. The software emulating this is called a hypervisor. Every VM interacts with the hypervisor through the operating system installed on the VM, which could be a typical desktop or laptop OS. There are many products that provide virtualised environments like Oracle VirtualBox, VMware Player, Parallel Desktop, Hyper-V, Citrix XenClient, etc.

Docker is based on the concept of containerisation. A container runs in an isolated partition inside the shared Linux kernel running on top of the hardware. There is no concept of emulation or a hypervisor in containerisation. Linux namespaces and cgroups enable Docker to run applications inside the container. In contrast to VMs, all that a container requires is enough of an operating system, supporting programs and libraries, and system resources to run a specific program. This means that, practically, you can put two to three times as many applications on a single server with containers than you can with a VM. In addition, with containers you can create a portable, consistent operating environment for development, testing and deployment. That's a winning triple whammy.

Why Docker instead of VMs?

- Faster delivery of applications.
- Portable and scales more easily.
- Get higher density and run more of a workload.
- Faster deployment leads to easier management.

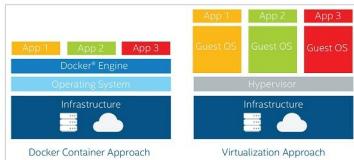


Figure 2: Containers vs VMs

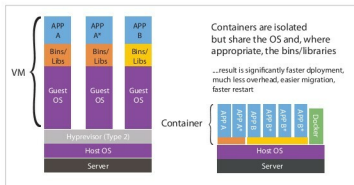


Figure 3: Containers are isolated

Docker features

- VE (Virtual Environments) based on LXC.
- Portable deployment across machines.
- *Versioning*: Docker includes Git-like capabilities for tracking versions of a container.
- *Component reuse*: It allows building or stacking of already created packages. You can create 'base images' and then run more machines based on the image.
- *Shared libraries*: There is a public repository with several images.
- Docker containers are very lightweight.

Who uses Docker and containers?

Many industries and companies have today shifted their infrastructure to containers or use containers in some other way.

The leading industries using Docker are energy, entertainment, financial, food services, life sciences, e-payments, retail, social networking, telecommunications, travel, healthcare, media, e-commerce, transportation, education and technology.

Some of the companies and organisations using Docker include The New York Times, PayPal, Business Insider, Cornell University, Indiana University, Splunk, The Washington Post, Swisscomm, GE, Groupon, Yandex, Uber, Shopify, Spotify, New Relic, Yelp, Quora, eBay, BBC News, and many more. There are many other companies planning to migrate their existing infrastructure to containers.

Integration of different tools

With the integration of various major tools available in the market now, Docker allows developers and IT operations teams to collaborate with each other to build more software